MATH 401. Homework #5. Sadullaev L.

2.2. Problem 5.

Let $a_n = \cos\left(\frac{n\pi}{2}\right)$. $\lim\limits_{n\to\infty} \sup a_n = ?$

$\lim\limits_{n\to\infty} \inf a_n = ?$

Solution:

$$\left.\begin{array}{ll}
n=1 & a_1 = \cos\frac{1\pi}{2} = 0. \\
n=2 & a_2 = \cos\frac{2\cdot\pi}{2} = \cos\pi = -1. \\
n=3 & a_3 = \cos\frac{3\pi}{2} = 0. \\
n=4 & a_4 = \cos\frac{4\pi}{2} = \cos 2\pi = 1. \\
n=5 & a_5 = \cos\frac{5\pi}{2} = 0 \\
n=6 & a_6 = \cos\frac{6\pi}{2} = -1 \\
\ldots\ldots &
\end{array}\right\} \Rightarrow \underset{m_k}{\underset{\parallel}{-1}} \le a_n \le \underset{M_k}{\underset{\parallel}{1}}.$$

$$\{M_k\} = \{a_k \mid k = 4n,\ n \in \mathbb{N}\} \qquad (1)$$

$$\{m_k\} = \{a_k \mid k = 4n+2,\ n \in \mathbb{N} \cup \{0\}\} \qquad (2)$$

$\lim\limits_{k\to\infty} M_k = \lim\limits_{n\to\infty} \sup a_n = 1$; $\lim\limits_{k\to\infty} m_k = \lim\limits_{n\to\infty} \inf a_n = -1$

$\blacksquare$

2.2. Problem 7.

Show that

(1) $\lim\limits_{n\to\infty} \sup(a_n+b_n) \leq \lim\limits_{n\to\infty} \sup a_n + \lim\limits_{n\to\infty} \sup b_n$

(1') $\lim\limits_{n\to\infty} \inf(a_n+b_n) \geq \lim\limits_{n\to\infty} \inf a_n + \lim\limits_{n\to\infty} \inf b_n$

Solution:

Let

(2) $L_n = \sup\{a_k \mid k \geq n\}$ (def. of sup.)

(3) $M_n = \sup\{b_k \mid k \geq n\}$ (def. of. sup.)

(4) $\left[\begin{array}{l} \sup(a_n+b_n) = \sup\{a_k+b_k \mid k\geq n\} \\ \lim\limits_{k\to\infty}\sup a_n = \lim\limits_{n\to\infty} L_n \; ; \; \lim\limits_{n\to\infty}\sup b_n = \lim\limits_{n\to\infty} M_n \end{array}\right.$ (def. of sup.)

Then

$$a_k + b_k \leq L_n + M_n \qquad \forall k \geq n \Rightarrow$$

$$\Rightarrow \lim_{n\to\infty}(a_k+b_k) \leq \lim_{n\to\infty}(L_n + M_n) \Rightarrow$$

[since $k \geq n$, $n \to \infty \to k \to \infty$], and by (4),

$$\Rightarrow \lim\sup(a_n+b_n) \leq \lim_{n\to\infty} L_n + \lim_{n\to\infty} M_n$$

$$\Rightarrow \lim_{n\to\infty} \sup(a_n + b_n) \leq \lim_{n\to\infty} \sup a_n + \lim_{n\to\infty} \sup b_n$$

by (4).

(4') Show that

$$\lim_{n\to\infty} \inf(a_n + b_n) \geq \lim_{n\to\infty} \inf a_n + \lim_{n\to\infty} \inf b_n$$

Let

(5) $l_n = \inf\{a_k \mid k \geq n\}$ & (def. of inf)

(6) $m_n = \inf\{b_k \mid k \geq n\}$ (def. of inf.)

Then

(7) $\lim_{n\to\infty} \inf a_n = \lim_{n\to\infty} l_n$; $\lim_{n\to\infty} \inf b_n = \lim_{n\to\infty} m_n$ $\left(\begin{array}{c}\text{def. of} \\ \text{lim inf}\end{array}\right)$

Then,

(8) $\inf(a_n + b_n) = \inf\{a_k + b_k \mid k \geq n\} \Rightarrow$

(9) $a_k + b_k \geq l_n + m_n \quad \forall k \geq n \Rightarrow$

(10) $\inf(a_n + b_n) \geq l_n + m_n$ by (8).

$$\liminf_{n\to\infty} (a_n + b_n) \geq \lim_{n\to\infty} (l_n + m_n) \Rightarrow$$

$$\Rightarrow \liminf_{n\to\infty} (a_n + b_n) \geq \lim_{n\to\infty} l_n + \lim_{n\to\infty} m_n \Rightarrow$$

(11) $$\liminf_{n\to\infty} (a_n + b_n) \geq \liminf_{n\to\infty} a_n + \liminf_{n\to\infty} b_n$$

by (7)

$\blacksquare$

2.2. Problem 9.

We shall see how we can define lim sup & lim inf for functions $f: \mathbb{R} \to \mathbb{R}$.

Let $a \in \mathbb{R}$, and define (note that we exclude $x = a$ in these definitions)

(1) $M_\varepsilon = \sup\{ f(x) \mid x \in (a-\varepsilon, a+\varepsilon), x \neq a \}$

(2) $m_\varepsilon = \inf\{ f(x) \mid x \in (a-\varepsilon, a+\varepsilon), x \neq a \}$

$\forall \varepsilon > 0$. (We allow $M_\varepsilon = \infty$, $m_\varepsilon = -\infty$).

(a) Show that M_ε decreases and m_ε increases as $\varepsilon \to 0$.

(b) Show that $\lim_{\varepsilon \to 0^+} M_\varepsilon$ & $\lim_{\varepsilon \to 0^+} m_\varepsilon$ exist

(We allow $M = \infty$, $m = -\infty$)

We know define

(3) $\lim_{x \to a} \sup f(x) = \lim_{\varepsilon \to 0^+} M_\varepsilon$ and

(4) $\lim_{x \to a} \inf f(x) = \lim_{\varepsilon \to 0^+} m_\varepsilon$

(c) Show that $\lim_{x \to a} f(x) = b$ if and only if

$$\lim_{x \to a} \sup f(x) = \lim_{x \to a} \inf f(x) = b.$$

(d) Find $\lim_{x \to 0} \inf \left(\sin \frac{1}{x}\right)$ and

$$\lim_{x \to 0} \sup \sin\left(\frac{1}{x}\right)$$

Solution:

(a) Let $f: \mathbb{R} \to \mathbb{R}$, and $a \in \mathbb{R}$.

Define

(1) $M_\varepsilon = \sup\{f(x) \mid x \in (a-\varepsilon, a+\varepsilon),\ x \neq a\}$

(2) $m_\varepsilon = \inf\{f(x) \mid x \in (a-\varepsilon, a+\varepsilon),\ x \neq a\}$, $\forall \varepsilon > 0$.

(Allowing $M_\varepsilon = \infty$ & $m_\varepsilon = -\infty$)

WTS: M_ε is $\downarrow$ & m_ε is $\uparrow$ as $\varepsilon \to 0$.

Let different (but close to zero, and approaching to 0 (zero) by decreasing) values of ε's in one set as following:
Since $\varepsilon_i \in \mathbb{R}$, we can order them:

(3) $\varepsilon = \{\varepsilon_1, \varepsilon_2, \varepsilon_3, \dots \varepsilon_n, \dots\}$ $\varepsilon_i > \varepsilon_{i+1}$

$\{\varepsilon\} \to 0$ as $n \to \infty$

Using the set ε, let

(4) $M_\varepsilon = \{M_1, M_2, M_3, \dots M_n, \dots\}$

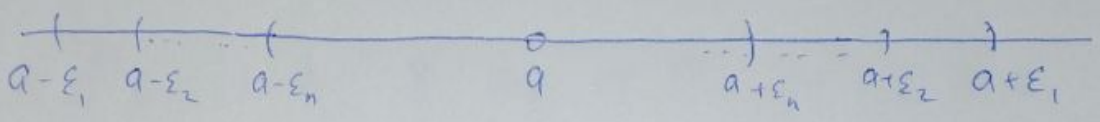
Since $f(x) \in \mathbb{R}$, we have the following supremums in $\mathbb{R}$ by completness:

(5) $M_i = \sup\{f(x) \mid x \in (a-\varepsilon_i, a+\varepsilon_i),\ x \neq a\} \in \mathbb{R}$

for $\varepsilon_i \in \mathcal{E}$.

Then by the decreasing order of the set $\mathcal{E}$ we have

(6) $(a-\varepsilon_{i+1},\ a+\varepsilon\ ) \subset (a-\varepsilon_i,\ a+\varepsilon_i),\ \forall i \in \mathbb{N}$



By (6), we can use the fact that we used in the proof of Prop. 4.2. $A \subset B \Rightarrow f(A) \subset f(B)$. $\Rightarrow$

$\Rightarrow f\big((a-\varepsilon_{i+1},\ a+\varepsilon_{i+1})\big) \subset f\big((a-\varepsilon_i,\ a+\varepsilon_i)\big)$, $\forall i \in \mathbb{N}$

that is

(7) $\{f(x) \mid x \in (a-\varepsilon_{i+1},\ a+\varepsilon_{i+1}),\ x \neq a\} \subset \{f(x) \mid x \in (a-\varepsilon_i,\ a+\varepsilon_i),\ x \neq a\} \Rightarrow$

$\Rightarrow$

$\sup\{f(x) \mid x \in (a-\varepsilon_{i+1},\ a+\varepsilon_{i+1}),\ x \neq a\} \underset{\downarrow}{\leq} \sup\{f(x) \mid x \in (a-\varepsilon_i,\ a+\varepsilon_i),\ x \neq a\}$

$\forall i \in \mathbb{N}$. with equality by (5). $\Rightarrow$

$\Rightarrow$

(9) $$M_{i+1} \leq M_i \quad \forall i \in \mathbb{N} \Rightarrow$$

$M_\varepsilon \subseteq \mathbb{R}$ is a decreasing sequence!

Similar statement can be made for inf.

Using the set ε

(3) $$\varepsilon = \{\varepsilon_1, \varepsilon_2, \dots \varepsilon_n, \dots\}$$

$$\varepsilon_i > \varepsilon_{i+1}, \quad \{\varepsilon\} \to 0 \text{ as } n \to \infty.$$

Let

(10) $$m_\varepsilon = \{m_1, m_2 \dots m_n \dots\}$$

since $f(x) \in \mathbb{R}$, we have the following infimums in $\mathbb{R}$ by completeness of $\mathbb{R}$.

$$m_i = \inf\{f(x) \mid x \in (a-\varepsilon_i, a+\varepsilon_i), x \neq a\} \in \mathbb{R}, \text{ for } \varepsilon_i \in \varepsilon.$$

Then by the decreasing order of the set ε, we have the nested intervals

(12) $$(a-\varepsilon_{i+1}, a+\varepsilon_{i+1}) \subset (a-\varepsilon_i, a+\varepsilon_i) \quad \forall i \in \mathbb{N}.$$

By (12), we can use the fact that $A \subset B \Rightarrow f(A) \subset f(B)$,

$\Rightarrow$

(13) $\{f(x) \mid x \in (a-\varepsilon_{i+1}, a+\varepsilon_{i+1}), x \neq a\} \subset \{f(x) \mid x \in (a-\varepsilon_i, a+\varepsilon_i), x \neq a\} \Rightarrow$ $\forall i \in \mathbb{N}$.

(14) $\Rightarrow \inf\{f(x) \mid x \in (a-\varepsilon_{i+1}, a+\varepsilon_{i+1}), x \neq a\} \geq \inf\{f(x) \mid x \in (a-\varepsilon_i, a+\varepsilon_i), x \neq a\}$

Since the right-hand side set in (13) contains the function values in the left-hand side $\Rightarrow$

$$m_{i+1} \geq m_i \quad \forall i \in \mathbb{N} \Rightarrow$$

$m_\varepsilon \subseteq \mathbb{R}$ is an increasing sequence!

(b) Show that $\lim\limits_{\varepsilon \to 0^+} M_\varepsilon$ & $\lim\limits_{\varepsilon \to 0^+} m_\varepsilon$ exist!

Solution:

By the result from part (a), $M_\varepsilon \subseteq \mathbb{R}$ is a decreasing sequence & $m_\varepsilon \subseteq \mathbb{R}$ is an increasing sequence.

Theor. 2.2.2.
Every monotone bounded sequence in $\mathbb{R}$ converges to a number in $\mathbb{R}$.

To use Thear 2.2.2, the only thing we need to show is $M_\varepsilon, m_\varepsilon$ are <u>bounded</u>, since they're monotone!

So by the notaiton in part (a)

$$M_\varepsilon = \{ M_1, M_2, M_3, \dots M_n, \dots \} \subseteq \mathbb{R} \Rightarrow$$

M_ε is not empty.

WTS: M_ε is bounded below that is

$\exists$ a number $M \in \mathbb{R}$ s.t. $M \leq M_i \quad \forall i \in \mathbb{N}$.

(6) Note that in the theorem of completeness, some times we shall allow $\sup A = \infty$ even when A is not bounded below. So still $\inf A$ exists allowing to receive $-\infty$. We can use this explotation of ∞ for M_ε sequence since we do not know how $f(x)$ behaves around $f(a)$, maybe $f(x)$ has a big jump into ∞ or $-\infty$ since $f(x)$ is not given to be continuous. So, we can pick $M = -\infty$ as a lower bound for M_ε, even $M = \inf M_\varepsilon = -\infty$ can be valid too.

So, $M = -\infty$ is a lower bound for M_ε which

means that by completeness principle $M_\varepsilon \subseteq \mathbb{R}$ is monotone decreasing, lower bounded (by $M = -\infty$), so $M_\varepsilon \to M$ in a similar proof manner in the proof of Theor. 2.2.2.

$M_\varepsilon \downarrow$ has a lower bound.

Now let

(7) $M = \inf M_\varepsilon = \inf \{ M_1, M_2, \dots M_n, \dots \}$

to show $\{M_\varepsilon\} \to M = -\infty$. $\Rightarrow$

We need to show $\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ s.t.

(8) $|M_n - M| < \varepsilon$.

As M is the greatest lower bound of M_ε, $M + \varepsilon$ can not be a lower bound, and hence there must be an M_N s.t.

(9) $$M_N < M + \varepsilon$$

since the sequence M_ε is $\downarrow$ (decreasing) by (5),

$$(10) \quad \overbrace{M \le \underbrace{M_n \le M_N}_{\text{decreasing}}}^{\text{by (5)}} \overbrace{< M+\varepsilon}^{\text{by (9)}} \qquad \underline{\forall n > N} \Rightarrow$$

pay attention to inclusion.

$$(11) \Rightarrow \quad M \le M_n < M+\varepsilon \qquad \underline{\forall n \ge N}.$$

$$\Rightarrow \quad \underline{M-\varepsilon} < M \le \underline{M_n} < \underline{M+\varepsilon} \qquad \forall n \ge N$$

$$\Rightarrow \quad M-\varepsilon < M_n < M+\varepsilon \quad , \forall n \ge N \Rightarrow$$

$$(12) \Rightarrow \quad |M_n - M| < \varepsilon \quad , \forall n \ge N. \Rightarrow$$

$$\{M_n\} \to M \quad \Rightarrow$$

$$(13) \quad \lim_{n\to\infty} M_n \underset{\text{as } n\to\infty \text{ by the def. of set } \varepsilon}{\overset{\text{note that } \varepsilon \to 0^+ \text{ by (3)}}{=\!=\!=}} \lim_{\varepsilon \to 0^+} M_\varepsilon = M.$$

$$\Rightarrow \lim_{n\to\infty} M_n = M \quad \text{exists.}$$

$\boxtimes$

Similar arguments above can be made for the increasing set $\{m_\varepsilon\}$ has bound $m = \infty$, and by Th. 2.2.2 $\{m_\varepsilon\} \to m$ exists.

$\boxtimes$

(c) Show that $\lim_{x \to a} f(x) = b$ iff

(15) $\lim_{x \to a} \sup f(x) = \lim_{x \to a} \inf f(x) = b.$

where

(16) $\lim_{x \to a} \sup f(x) = \lim_{\varepsilon \to 0^+} M_\varepsilon$ &

(17) $\lim_{x \to a} \inf f(x) = \lim_{\varepsilon \to 0^+} m_\varepsilon$

Solution:

($\Rightarrow$) Let (16) & (17) and assume

(18) $\lim_{x \to a} f(x) = b \Rightarrow$

$\forall \varepsilon > 0, \exists \delta > 0$ s.t.

(19) $|f(x) - f(b)| < \varepsilon$ whenever $|x-a| < \delta$ holds.

$\Rightarrow$ $-\varepsilon < f(x) - b < \varepsilon$ whenever $|x-a| < \delta \Rightarrow$

(20) $b - \varepsilon < f(x) < b + \varepsilon$ whenever $|x-a| < \delta$.

Recall

note matches to the assumption whenever part.

(21) $M_\delta = \sup\{f(x) \mid |x-a| < \delta,\ x \neq a\}$

(22) $m_\delta = \inf\{f(x) \mid |x-a| < \delta,\ x \neq a\}$

Continuing (20):

(20') $b - \varepsilon < m_\delta \leq f(x) \leq M_\delta < b + \varepsilon$ whenever $|x-a| < \delta$, $x \neq a$

$\Rightarrow b - \varepsilon < m_\delta < b + \varepsilon$ whenever $|x-a| < \delta$, $x \neq a$

$\Rightarrow |m_\delta - b| < \varepsilon \Rightarrow \forall \varepsilon > 0 \Rightarrow$

(23) $\lim\limits_{\delta \to 0^+} m_\delta = \lim\limits_{x \to a} \inf f(x) = b.$

Similarly, by (20') above:

$b - \varepsilon < M_\delta < b + \varepsilon \Rightarrow \forall \varepsilon > 0$

(24) $\Rightarrow \lim\limits_{\delta \to 0^+} M_\delta = \lim\limits_{x \to a} \sup f(x) = b$

whenever $|x-a| < \delta$, $x \neq a$.

$(\Longleftarrow)$

(25) Assume $\lim_{\delta \to 0^+} M_\delta = \lim_{x \to a} \sup f(x) = b$ &

(26) $\lim_{\delta \to 0^+} m_\delta = \liminf_{x \to a} f(x) = b$ whenever $|x-a| < \delta$, $x \neq a$ (def. of M_δ)

$\Rightarrow \quad m_\delta \leq f(x) \leq M_\delta$ whenever $|x-a| < \delta$, $x \neq a$.

$\Rightarrow \quad f(x) \to b$ when $|x-a| < \delta$ $x \neq a$. $\Rightarrow$

$\Rightarrow \lim_{x \to a} f(x) = b.$ $\blacksquare$

(d) Find

(i) $\liminf_{x \to 0} \sin \frac{1}{x} = ?$

(ii) $\limsup_{x \to 0} \sin \frac{1}{x} = ?$

Solution:

$$-1 \leq \sin \frac{1}{x} \leq 1$$

So,

(ii)

(1) $\sin\frac{1}{x} = 1 \Rightarrow \frac{1}{x} = \frac{\pi}{2} + 2\pi n \Rightarrow$

$$\{x_n\} = \left\{\frac{2}{\pi(4n+1)}\right\} \to 0 \quad \text{as given.} \Rightarrow$$

(2) $\Rightarrow \limsup_{x\to 0} \sin\left(\frac{1}{x}\right) = \lim_{n\to\infty} M_n = \lim_{n\to\infty} \sin\left(\frac{1}{\frac{2}{\pi(4n+1)}}\right)$

$= \lim_{n\to\infty} \sin\left(\frac{\pi}{2} + 2\pi n\right) = \lim_{n\to\infty} \sin\frac{\pi}{2} = 1 \Rightarrow$

(3) $\limsup_{x\to 0} \sin\left(\frac{1}{x}\right) = 1.$

(i) similarly,

(4) $\sin\frac{1}{x} = -1 \Rightarrow \frac{1}{x} = \frac{3\pi}{2} + 2\pi n \Rightarrow x = \frac{2}{3\pi + 4\pi n} \Rightarrow$

$$\{x_n\} = \left\{\frac{2}{\pi(4n+3)}\right\} \to 0 \quad \text{as } n\to\infty. \quad \text{So}$$

(5) $\liminf_{x\to 0} \sin\left(\frac{1}{x}\right) = \lim_{n\to\infty} m_n = \lim_{n\to\infty} \sin\left(\frac{1}{\frac{2}{\pi(4n+3)}}\right) =$

$= \lim_{n\to\infty} \sin\left(\frac{3\pi}{2} + 2\pi n\right) = \lim_{n\to\infty} \sin\frac{3\pi}{2} = -1.$

$\Rightarrow$

(6) $\lim_{x \to 0} \inf \sin\left(\frac{1}{x}\right) = -1$

We can use the result from part (c) that (5) and (6) are not equal, so we can conclude that

$$\lim_{x \to 0} \sin\left(\frac{1}{x}\right) = \text{DNE (Does not exist)}$$

$\blacksquare$